Rational equations

1 Use technology to solve the following equations. Round your answer to two decimal places where necessary.

a $\frac{1}{x} = 5$

b $-\frac{1}{x} = -5x$

c $\frac{1}{x-8} = -\frac{1}{x+1}$

d $\frac{1}{2x^2} = 5$

e $\frac{8x-22}{x-3} = \frac{3x-25}{x-9}$

f $\frac{-5x^2-40x-81}{x^2+8x+16} = \frac{-5x-19}{x+4}$

g $\frac{10}{x^2+2x-15} = \frac{x^2+10}{5}$

h $\frac{x^2+x-12}{x-5} = \frac{2x+3}{x+4}$

2 Consider the equation $\frac{5}{x+3} + \frac{4}{x+4} = \frac{5}{x^2+7x+12}$.

a Solve for x.

b Explain whether your answer from part (a) is a valid solution.

3 For each of the following equations, determine whether $x = 4$ is a valid solution, an extraneous solution, or not a solution at all:

a $\frac{x+3}{x-4} = 0$

b $\frac{x-4}{x+3} = 0$

c $\frac{(x-4)(x+3)}{x-4} = 0$

d $\frac{x-4}{x-4} = 0$

e $\frac{(x-4)^2}{x-4} = 0$

f $\frac{x}{4} - \frac{4}{x} = 0$

4 Find the valid solutions to the following equations:

a $\frac{1}{x-4} + \frac{1}{x+9} = 0$

b $\frac{x}{x-2} + \frac{1}{x+2} = \frac{8}{x^2-4}$

c $\frac{9}{x+4} - \frac{5}{x-4} = \frac{2x}{x^2-16}$

d $\frac{2}{3x+5} + \frac{2}{3x-5} = \frac{5x}{9x^2-25}$

e $\frac{3}{x^2-2x} - \frac{5}{x^2-4} = 0$

f $\frac{5}{x^2-2x} - \frac{3}{x^2-4x} = 0$

g $1 - \frac{7}{x} + \frac{10}{x^2} = 0$

h $1 + \frac{9}{x} = -\frac{20}{x^2}$

i $\frac{3x}{3-x} + \frac{3}{x} - 5 = 0$

j $\frac{2x}{x-8} + \frac{6}{x+8} = \frac{4}{64-x^2}$

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5 Find the valid solutions to the following equations:

a $2x^{-1} + 5(2x-5)^{-1} = 4(2x-5)^{-1}$

b $3x^{-1} + 2(5x-4)^{-1} = 5(5x-4)^{-1}$

c $x(x-9)^{-1} + x(x+9)^{-1} = 32\left(x^2-81\right)^{-1}$

d $x(x-7)^{-1} + x(x+7)^{-1} = 18\left(x^2-49\right)^{-1}$

6 Find all the complex solutions to $x^{-4} + 8x^{-2} - 9 = 0$.

Graphical solutions to rational equations

7 For each of the following functions, use technology to find:

i The number of x-intercepts.

ii The number of y-intercepts.

iii The number of horizontal asymptotes.

iv The number of vertical asymptotes.

a $f(x) = \dfrac{x-5}{x-3}$

b $f(x) = \dfrac{x^2+3x-10}{x+7}$

c $f(x) = \dfrac{x+2}{x^2+x-6}$

d $f(x) = \dfrac{x^2+5x-14}{x^2-3x-18}$

8 Sketch the graph of the following functions:

a $y = \dfrac{4(5-x)}{x-4}$

b $f(x) = \dfrac{x+5}{x-9}$

c $y = \dfrac{4}{(x-3)^2}$

d $y = \dfrac{x^2}{(x-6)(x+5)}$

e $y = \dfrac{-10(x+2)}{(x+5)(x-8)}$

f $f(x) = \dfrac{x-2}{x^2-14x+48}$

g $f(x) = \dfrac{x^2-4x-45}{x^2-6x-16}$

h $f(x) = \dfrac{x^2-x-6}{x+6}$

9 Consider the rational equation $\dfrac{3}{x+4} = -\dfrac{1}{x-3}$.

a Sketch the graph of $y = \dfrac{3}{x+4}$.

b Sketch the graph of $y = -\dfrac{1}{x-3}$ on the same coordinate plane.

c State the solution to the rational equation.

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10 Consider the rational equation $\frac{4}{x-3} - \frac{6}{1-x} = 0$.

a Sketch the graph of $y = \frac{4}{x-3}$.

b Sketch the graph of $y = \frac{6}{1-x}$ on the same coordinate plane.

c Find the solution to the rational equation.

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